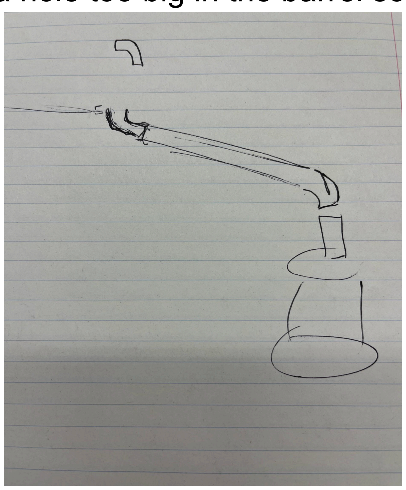
- Gather all materials at the coop that we know we needed so far.
- Patching original gutter/redirecting water to our new gutter disposal
- Cut estimated hole in the gutter for our new pipe
- Connect piping pieces together with silicone glue and setting it out to dry
    - First, gluing one elbow piece to the main gutter pipe which then we glued another elbow piece on the other side facing down. Finally we glued a small gutter pipe piece to the downward facing elbow that will directly connect to the barrel.
- Drilling 2 holes in the barrel, one that will hold the spicket and the other will hold the (pvc) pipe that will go through the coop connected to the pvc pipe in the coop.
- Working on the pvc pipe that will be inside the chicken coop that will provide water to the chickens: we first used some math to make the 4 holes we would drill for the nipples evenly apart, once solved we drilled the 4 holes that the nipples will go into (The Chicken Chick states 1 nipple per 9 hens, we have 11 chickens but figured an extra 2 wont hurt to have), we were able to securely get 2 nipples into the drilled holes but the others needed to be put in with silicone glue so they wouldn't fall out and no water would leak.
- Oops, we drilled a hole too big in the barrel so now we need to plug it with



expanding foam.

a. We used a plethora of tools throughout our project.
    - Drils
    - Shovels
    - Lots of silicone glue
    - Foaming glue
    - Pocket knives
    - Gutter pipes
    - Pvc pipes
    - Drinking spigots for the chickens
    - Spigot
    - Rain barrel
    - Zip ties
    - Gitter filter
    - Recycled cans
    - Recycled plastic bottles

b. There were a good amount of different techniques that we used as a group. Talking through our ideas with eachother and drawing out what we are thinking proved to be the most useful for us.
c. Our prototype was actually pretty effective, we built out our ideas with reused materials and discussed how different parts of it may or may not work as well as we had thought.